Sprint Plan — Online Examination Portal

Course: UE24CS341A — Software Engineering, PES University, Dept. of CSE
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1. Backlog

The backlog is the GitHub Project **Online Exam Portal - Backlog**, linked to this repository. It holds **one issue per requirement in `Documents/SRS.md`** — 30 functional (`FR-001`…`FR-030`) and 14 non-functional (`NFR-001`…`NFR-014`) — each with story points, an assignee, a priority taken from the SRS and a sprint.

Story points use the Fibonacci scale 1, 2, 3 (5 and 8 are reserved for items too large to finish inside a one-week sprint; none were needed): 1 = configuration or a single check, 2 = one screen or one rule, 3 = a rule with timing, concurrency or several screens.

Phase	Dates	Purpose
Backlog creation	by 12 Oct	Issues created from the SRS, story points and assignees set
Dry run	12–16 Oct	Practise the workflow: branch → PR → review → merge, board automation
Backlog freeze	19 Oct	Last day entries may change
Sprint 1	**19–23 Oct**	Thin end-to-end slice: Faculty sets an exam, a Student takes it, it is scored automatically
Sprint 2	**26–30 Oct**	Robust attempt, results publication, integrity and audit, NFR verification; development freeze

2. Sprint 1 — 19 to 23 October

Sprint goal: *A Faculty member creates a course, enrolls a Student, writes questions, builds and schedules an examination; the Student signs in, takes it under a server timer, and receives an automatically computed score.* The one-minute Sprint 1 video records exactly this path.

Req ID	Backlog item	Use case	Owner	Points
FR-001	Student self-registration	UC-01	Kartik	2
FR-002	Email and password sign-in with session	UC-02	Kartik	2
FR-003	One role per account, enforced on every screen and endpoint	UC-02	Kshitij	3
FR-004	Course create / edit / archive, assign Faculty	UC-03	Komal	2
FR-005	Enroll Students into a Course	UC-04	Komal	2
FR-006	Show and allow only exams of enrolled Courses	UC-04	Komal	3
FR-007	Single-correct MCQ authoring (2–6 options)	UC-05	Kartik	3
FR-008	Mandatory question tags and marks	UC-05	Kartik	2
FR-010	Create an Examination with its policy values	UC-06	Komal	3
FR-011	Add bank questions to an Examination	UC-06	Komal	2
FR-012	Draft / Published examination state	UC-06	Komal	2
FR-013	Schedule the examination window	UC-07	Kartik	2
FR-015	Attempt eligibility: window, enrollment, one attempt	UC-08	Kshitij	3
FR-016	Server-stamped attempt record before the paper	UC-08	Kshitij	3
FR-017	Countdown from the server clock	UC-08	Kartik	3
FR-020	Automatic submission at expiry	UC-10	Kshitij	3
FR-022	Automatic evaluation of every submission	UC-11	Kshitij	3
FR-023	Positive / negative / zero marking	UC-11	Kshitij	2
FR-029	No correct-answer data in any Student payload	UC-13	Kartik	2
NFR-006	Feature branches, PR by assignee, review by another member	—	Komal	1
NFR-011	Salted adaptive password hashing	—	Kartik	1
NFR-013	CSRF tokens, ORM-only queries, generic error pages	—	Komal	2
NFR-014	Secrets from environment variables, none in the repo	—	Komal	1

Sprint 1 capacity: 52 points — Kshitij 17 · Kartik 17 · Komal 18.

Also in Sprint 1 (course requirement, not a backlog item): Django project skeleton and a GitHub Actions workflow that runs `python manage.py test` on every push and pull request.

3. Sprint 2 — 26 to 30 October

Sprint goal: *The attempt survives interruptions, Students see published scorecards, focus loss and every change are recorded, and the non-functional requirements are verified.* The two-minute Sprint 2 video shows the full lifecycle including resume, publication and the audit trail.

Req ID	Backlog item	Use case	Owner	Points
FR-009	Version a question instead of editing it once in use	UC-05	Kshitij	3
FR-014	Student dashboard: Upcoming / Live / Completed	UC-07	Kartik	2
FR-018	One question at a time with the 4-state palette	UC-08	Kartik	3
FR-019	Autosave every response within 5 s	UC-09	Kshitij	3
FR-021	Resume an interrupted attempt	UC-09	Kshitij	3
FR-024	Total, percentage and Pass / Fail	UC-11	Kshitij	2
FR-025	Withhold results until published	UC-12	Komal	2
FR-026	Publish and unpublish results	UC-12	Komal	2
FR-027	Student scorecard with per-question marks	UC-12	Kartik	2
FR-028	Record focus-loss events with server time	UC-13	Kartik	2
FR-030	Append-only audit trail	UC-13	Komal	3
NFR-001	95% of requests ≤ 2 s at 200 concurrent attempts	—	Komal	3
NFR-002	Evaluate a 100-question attempt ≤ 5 s	—	Kshitij	1
NFR-003	200 concurrent attempts, 50,000-question bank	—	Komal	2
NFR-004	Service back ≤ 60 s after a restart, attempts resumable	—	Kshitij	2
NFR-005	Chrome / Edge / Firefox, usable at 360 px	—	Kartik	2
NFR-007	Every FR linked to a test case	—	Komal	1
NFR-008	DPDP Act consent and erasure	—	Kartik	2
NFR-009	WCAG 2.1 AA on the attempt flow	—	Kartik	2
NFR-010	Policy values configurable, not hard-coded	—	Kshitij	1
NFR-012	Lock account after 5 failed sign-ins	—	Komal	2

Sprint 2 capacity: 45 points — Kshitij 15 · Kartik 15 · Komal 15.

4. How each sprint runs

When	Event
Monday (19 Oct, 26 Oct)	**Sprint planning** — confirm the items above, split them into task sub-issues, move them to *Todo*
Tuesday – Thursday	Work in branches; each item's owner raises the PR that closes it (`Closes #<issue>`); another member reviews and merges
Daily	Short stand-up in the team group: done yesterday, doing today, blocked by
Friday (23 Oct, 30 Oct)	**Sprint review** — record the sprint video and upload it to `videos/`; **retrospective** notes appended to this file

Definition of done for a backlog item:

1. The code is merged into `main` through a pull request raised by the item's owner and reviewed by another team member (NFR-006).
2. The GitHub Actions check is green on that pull request.
3. The item's test cases in `Documents/Test_Plan.md` have been executed and their *Actual Result* and *Test Result* columns filled in.
4. The issue is closed by the merge and its board card is in *Done*.

Deviations. Any departure from the SRS, SAD or Test Plan is recorded in `Documents/Deviations.md` when it happens — what changed, why, and which requirement it affects — for explanation at the final demo. Development is **frozen** at the end of Sprint 2 (30 Oct); after that only documentation changes.

5. Retrospectives

To be filled in at the end of each sprint.